

AI511/MM505 Linear Algebra with Applications

Lecture 5 – Euclidean Vector Spaces, Linear Transformations

Vaidotas Characiejus

September 21, 2026

Department of Mathematics and Computer Science
University of Southern Denmark

Outline

Euclidean vector spaces

Vectors

Bases

Linear transformations

Euclidean vector spaces

Euclidean vector spaces

Vectors

Vectors

- $1 \times n$ - and $m \times 1$ -matrices play a special role and are called *row*- and *column* *vectors*, respectively.
- When we say that v is a vector, we typically do not specify whether it is a column vector (i.e., $\mathbb{R}^{n \times 1}$) or a row vector (i.e., $\mathbb{R}^{1 \times n}$)
- When we think of vectors geometrically, this difference is often unimportant and we simply write

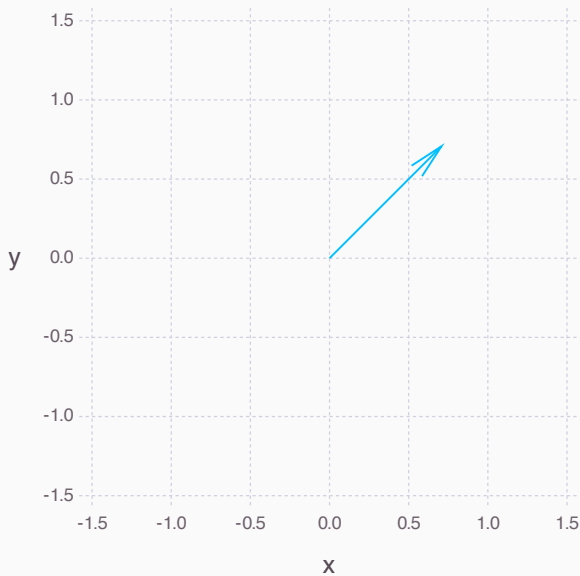
$$v = (v_1, \dots, v_n) \in \mathbb{R}^n.$$

- However, when we perform matrix multiplication, the difference between row vectors and column vectors is important
- When the shape of a vector matters and we give no indication otherwise, vectors are assumed to be column vectors.

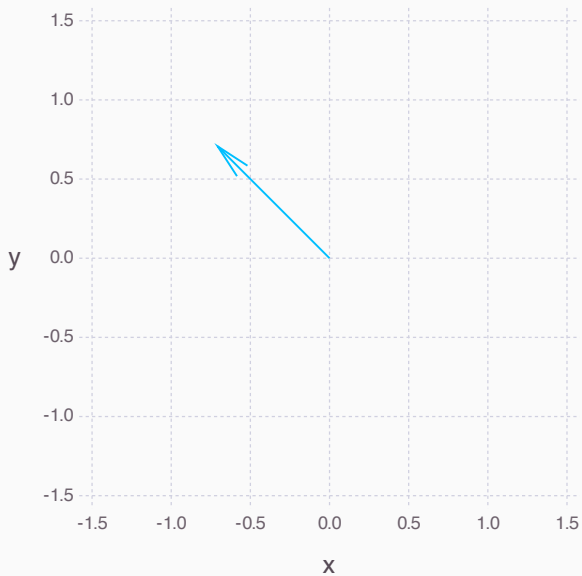
Vectors and geometry

- We can add vectors and multiply them by scalars, just as we do with matrices.
- These operations have a natural geometric interpretation in $\mathbb{R}^2$.
- A geometric viewpoint provides an intuitive way to understand many of the fundamental ideas in linear algebra.
- Recommended to watch *Vectors, what even are they?* of 3Blue1Brown.
- How do we visualise vectors and their operations geometrically?

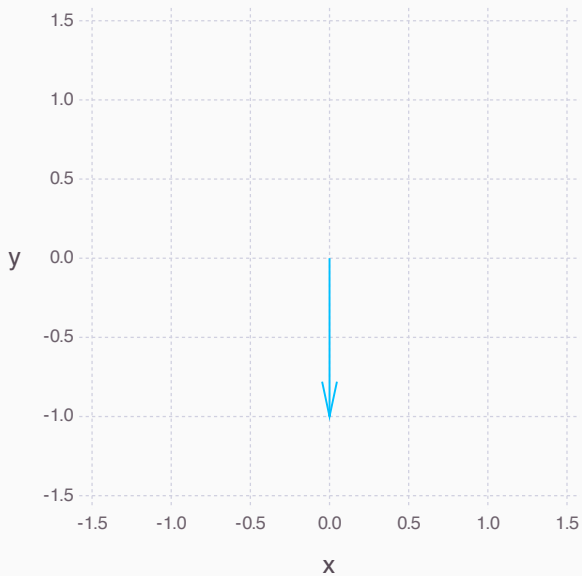
Single vector in $\mathbb{R}^2$



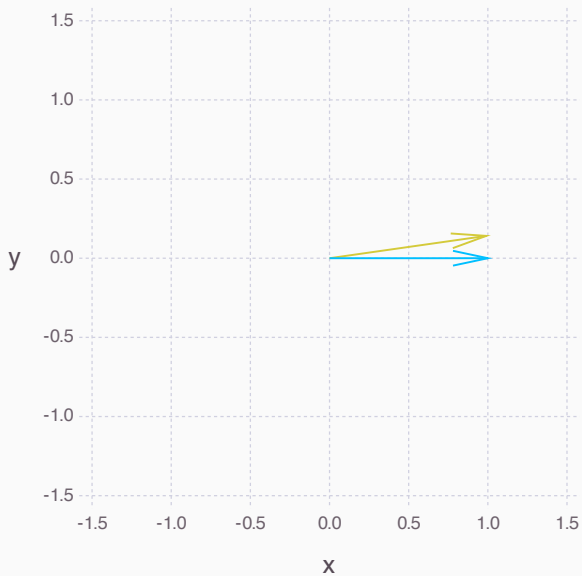
Single vector in $\mathbb{R}^2$



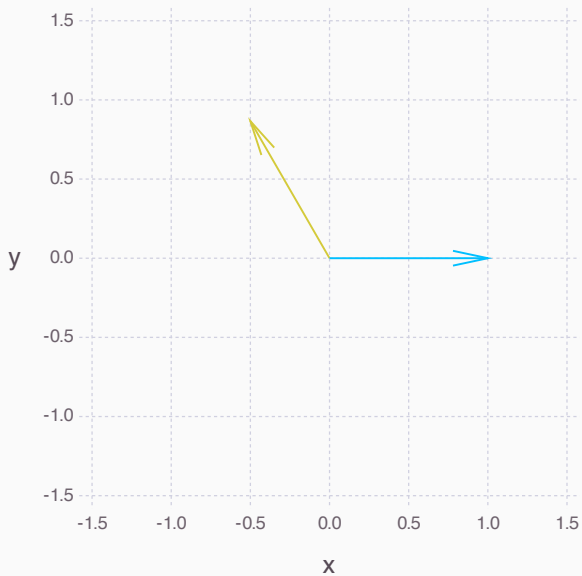
Single vector in $\mathbb{R}^2$



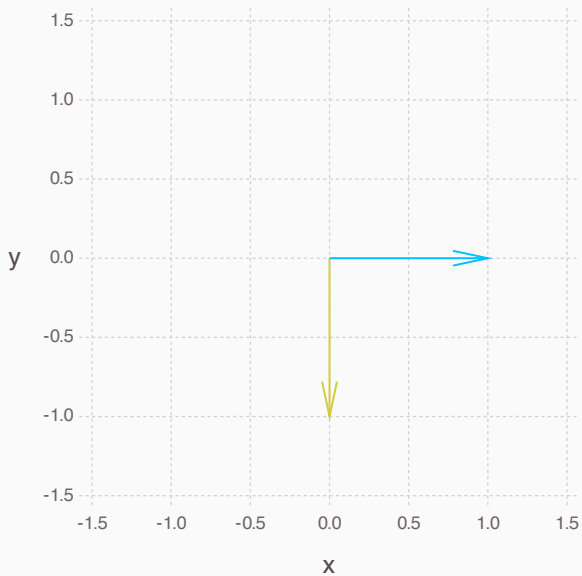
Two vectors in $\mathbb{R}^2$



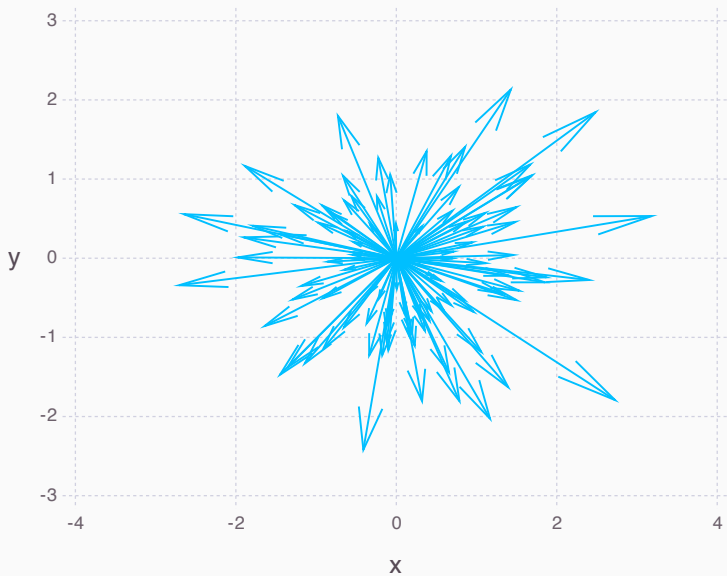
Two vectors in $\mathbb{R}^2$



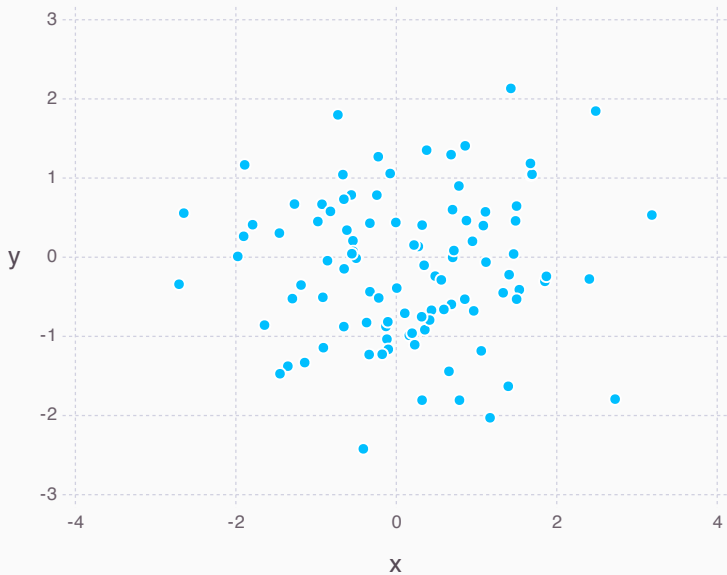
Two vectors in $\mathbb{R}^2$



Many vectors in $\mathbb{R}^2$



Many vectors in $\mathbb{R}^2$



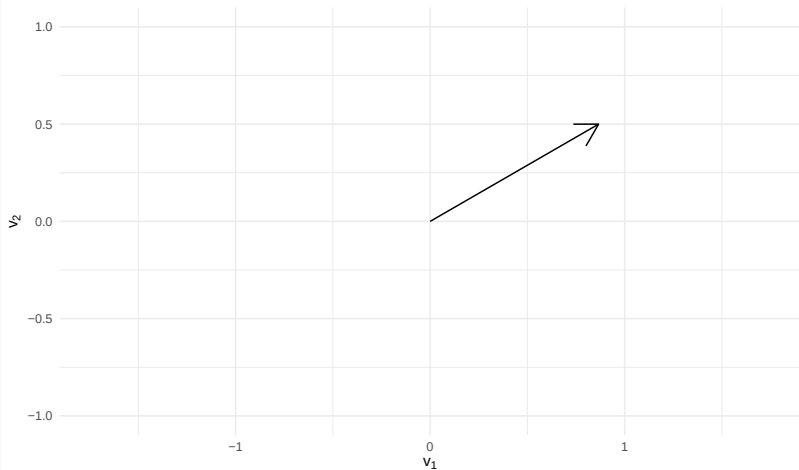
Scalar multiplication

Definition 1.1.2 of Johnston (2021)

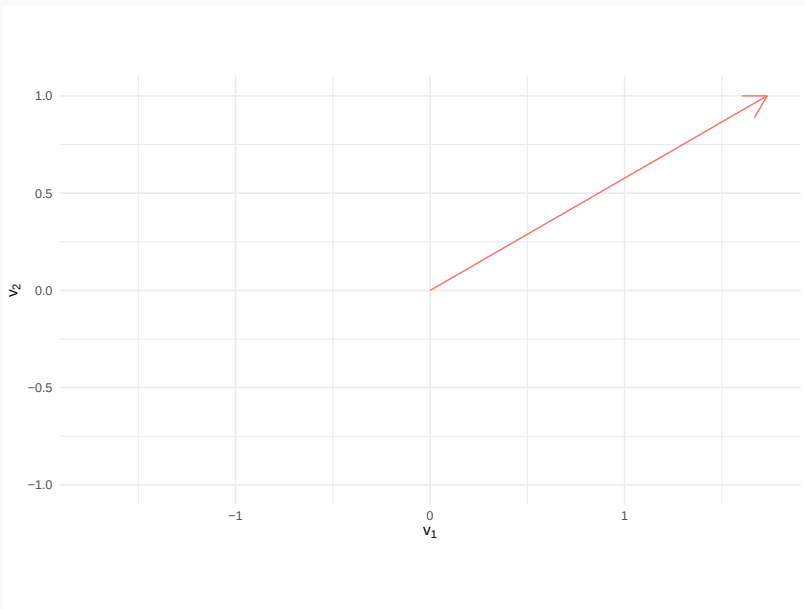
Suppose $v = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$ is a vector and $c \in \mathbb{R}$ is a scalar. Then their *scalar multiplication*, denoted by cv , is the vector

$$cv = (cv_1, cv_2, \dots, cv_n).$$

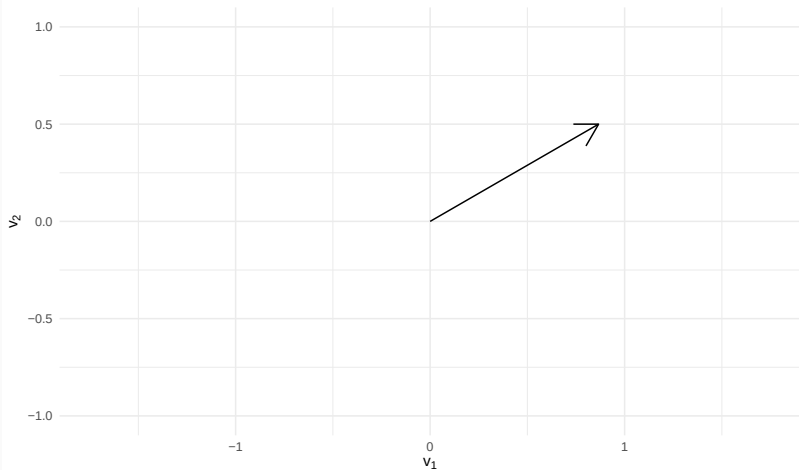
Scaling vectors in $\mathbb{R}^2$: $v = (\cos(\pi/6), \sin(\pi/6))$



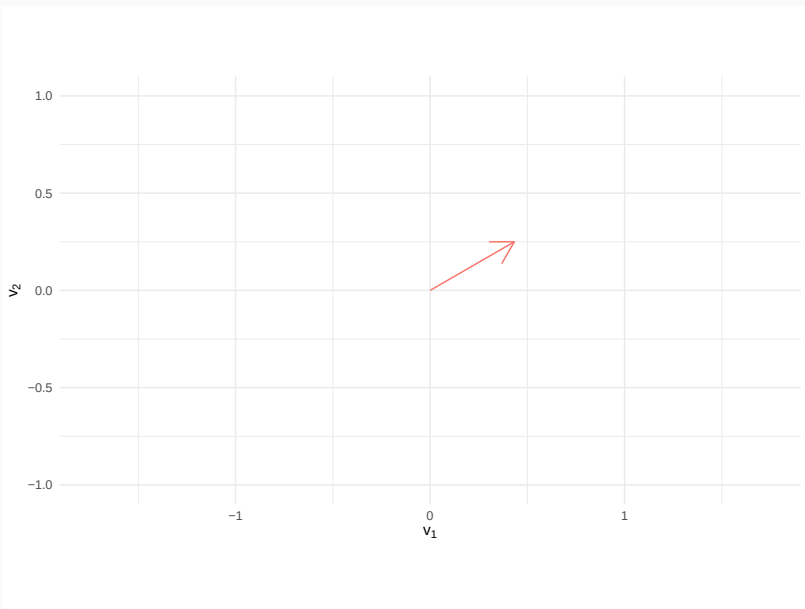
Scaling vectors in $\mathbb{R}^2$: $2v = 2(\cos(\pi/6), \sin(\pi/6))$



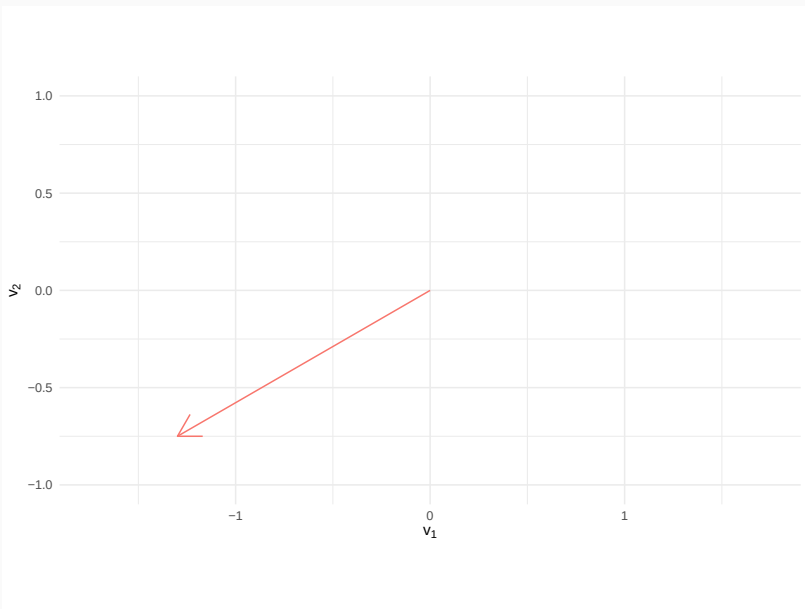
Scaling vectors in $\mathbb{R}^2$: $v = (\cos(\pi/6), \sin(\pi/6))$



Scaling vectors in $\mathbb{R}^2$: $.5v = .5(\cos(\pi/6), \sin(\pi/6))$



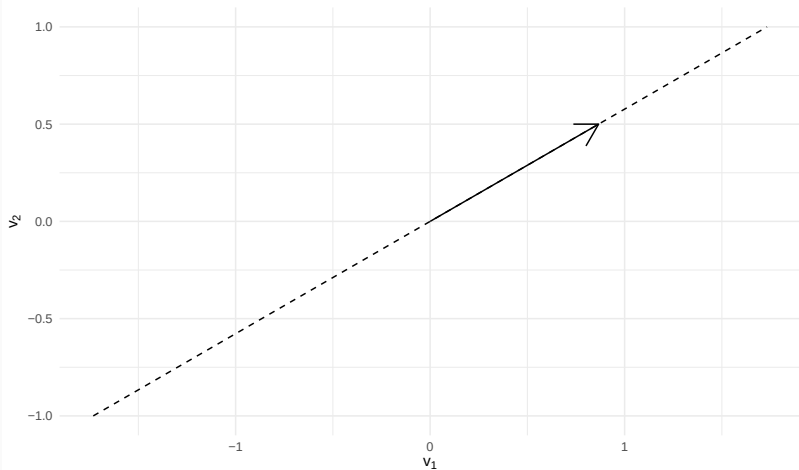
Scaling vectors in $\mathbb{R}^2$: $-1.5v = -1.5(\cos(\pi/6), \sin(\pi/6))$



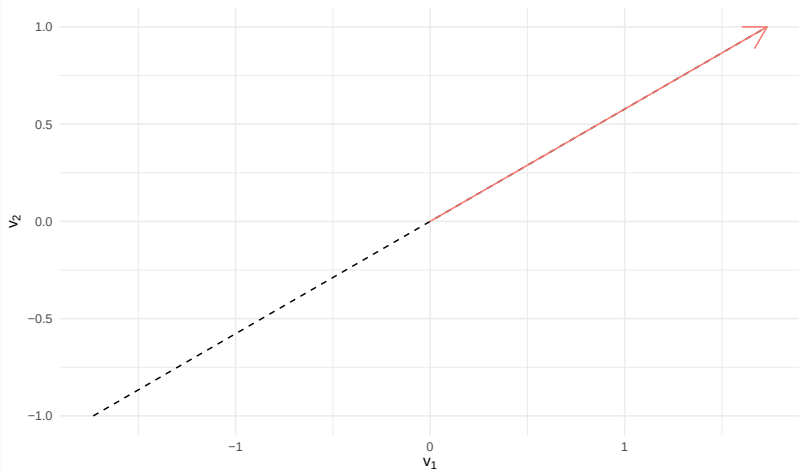
Scaling vectors in $\mathbb{R}^2$: $-.25v = -.25(\cos(\pi/6), \sin(\pi/6))$



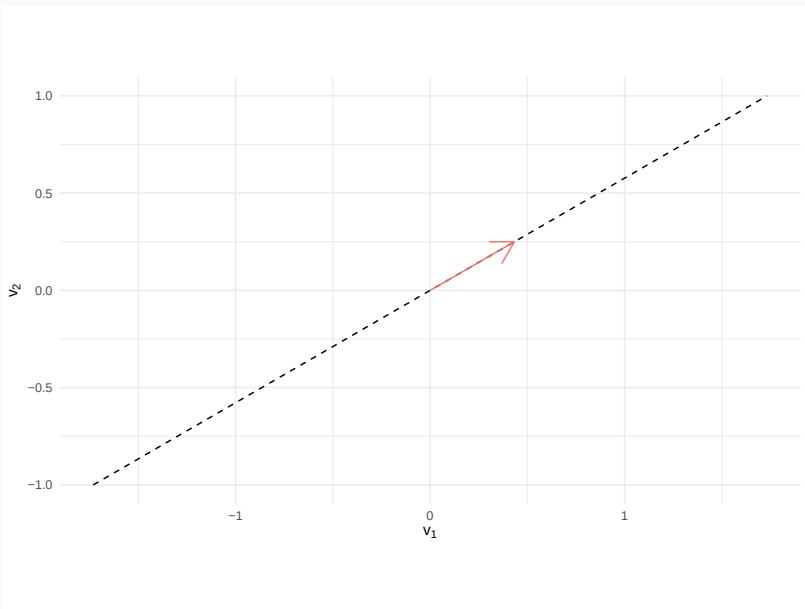
Scaling vectors in $\mathbb{R}^2$: $v = (\cos(\pi/6), \sin(\pi/6))$



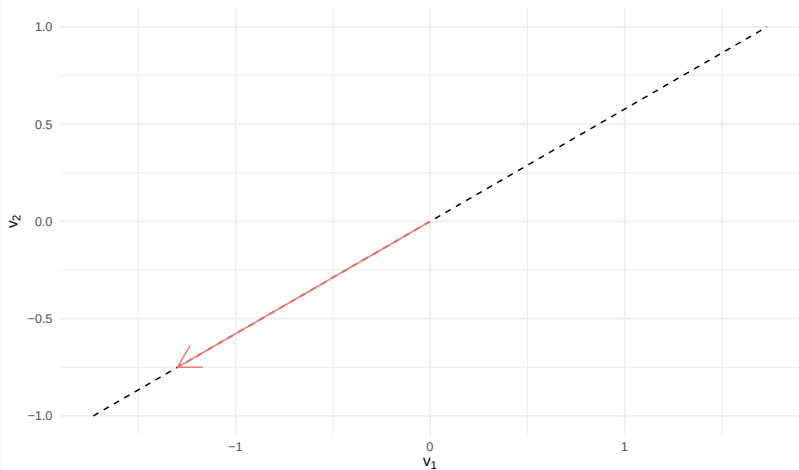
Scaling vectors in $\mathbb{R}^2$: $2v = 2(\cos(\pi/6), \sin(\pi/6))$



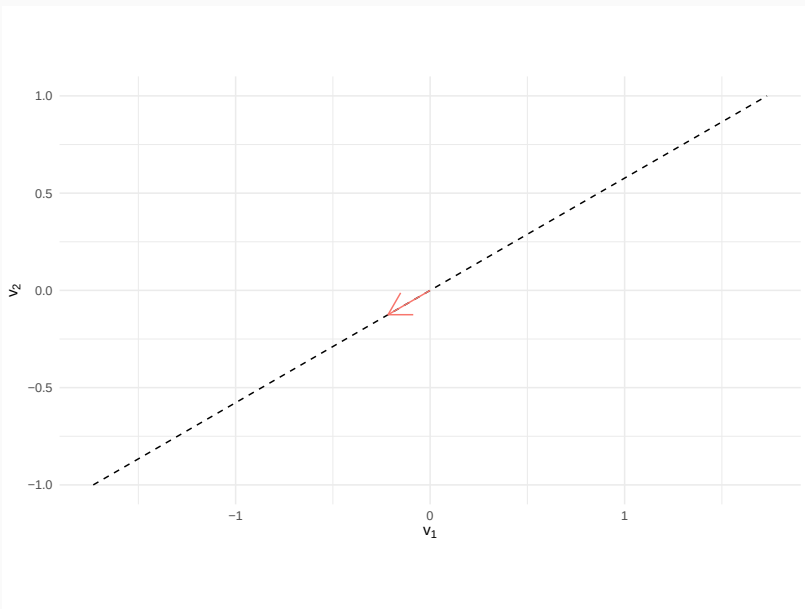
Scaling vectors in $\mathbb{R}^2$: $.5v = .5(\cos(\pi/6), \sin(\pi/6))$



Scaling vectors in $\mathbb{R}^2$: $-1.5v = -1.5(\cos(\pi/6), \sin(\pi/6))$



Scaling vectors in $\mathbb{R}^2$: $-.25v = -.25(\cos(\pi/6), \sin(\pi/6))$

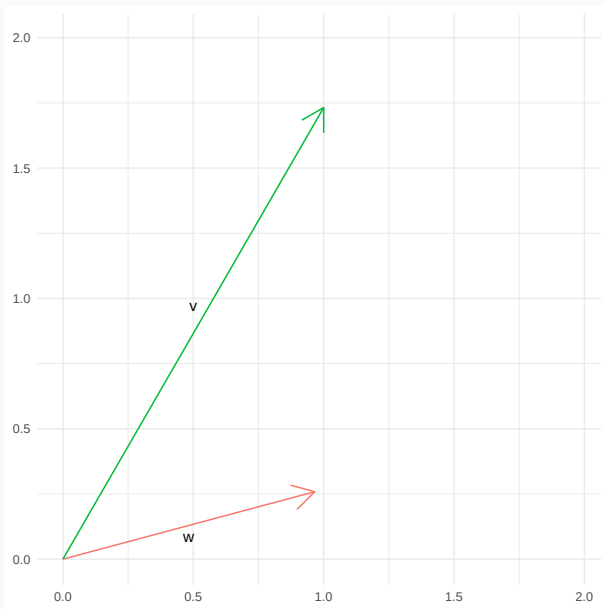


Definition 1.1.1 of Johnston (2021)

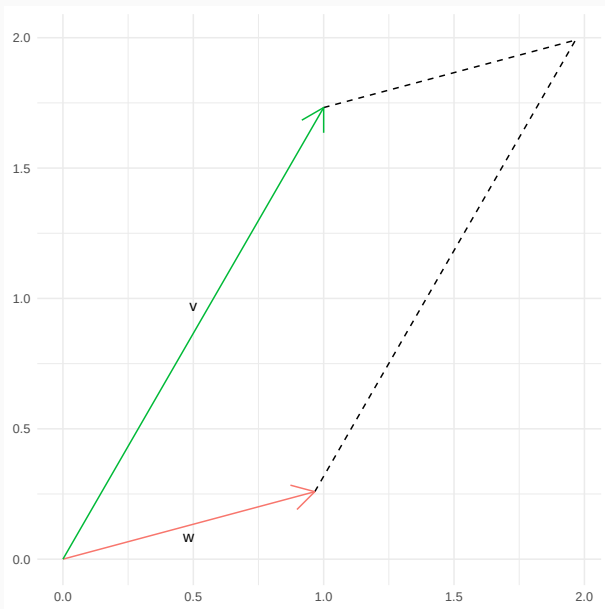
Suppose $v = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$ and $w = (w_1, w_2, \dots, w_n) \in \mathbb{R}^n$ are vectors. Then their *sum*, denoted by $v + w$, is the vector

$$v + w = (v_1 + w_1, v_2 + w_2, \dots, v_n + w_n).$$

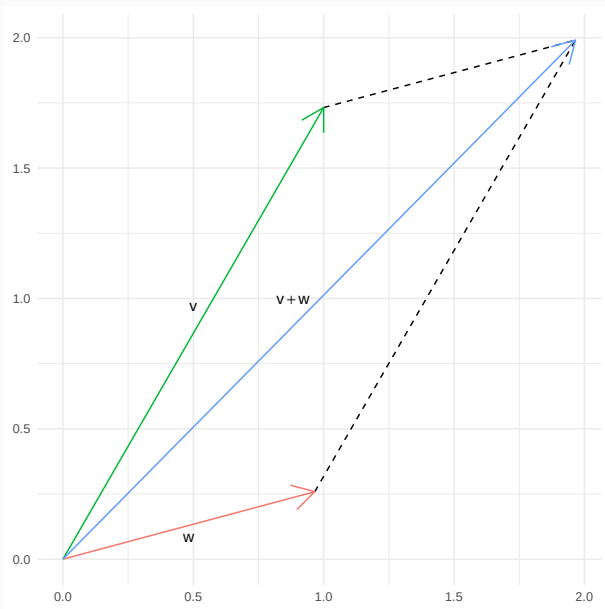
Vector addition in $\mathbb{R}^2$



Vector addition in $\mathbb{R}^2$



Vector addition in $\mathbb{R}^2$



Euclidean vector space

- With vector addition and scalar multiplication, we obtain a vector space (or a linear space).
- A vector space tells us how vectors can be added and scaled, but not how to measure their length, distance, or angle.
- Introducing a dot product (or, more generally, an inner product) adds this geometric structure, giving us a Euclidean vector space.

Dot product

Definition 1.2.1 of Johnston (2021)

Suppose $v = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$ and $w = (w_1, w_2, \dots, w_n) \in \mathbb{R}^n$. Then their *dot product*, denoted by $v \cdot w$, is the quantity

$$v \cdot w := v_1 w_1 + v_2 w_2 + \dots v_n w_n.$$

- Observe that

$$v \cdot w = v^T w = w^T v. \tag{1}$$

- The dot product can be interpreted geometrically as roughly measuring the amount of overlap between v and w .

The angle between two vectors on the unit circle in $\mathbb{R}^2$

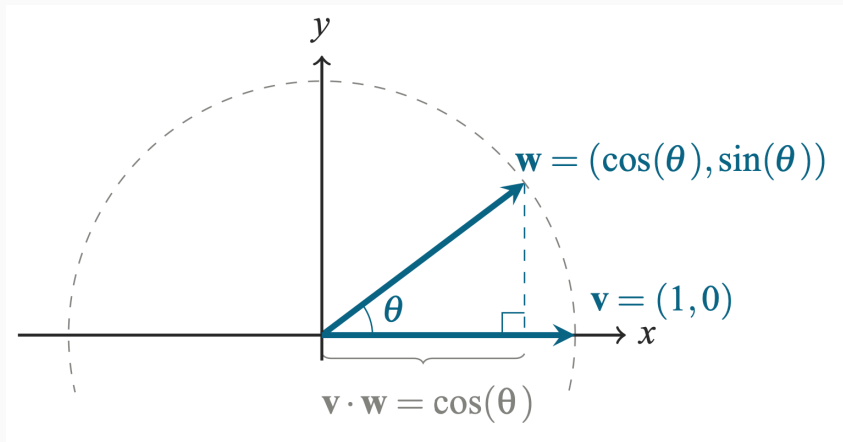


Figure 1.7 of Johnston (2021)

Orthogonal vectors

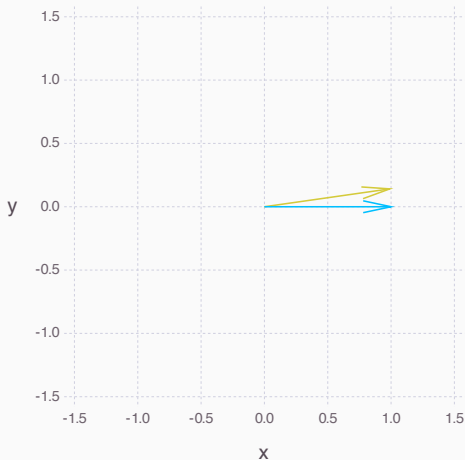
Definition 1.2.4 of Johnston (2021)

Two vectors $v, w \in \mathbb{R}^n$ are said to be *orthogonal* if

$$v \cdot w = 0.$$

What is the angle between two orthogonal vectors?

Vectors on the unit circle in $\mathbb{R}^2$: true or false?



(a) $v \cdot w > 0$;

(b) $v \cdot w < 0$;

(c) $v \cdot w = 0$;

(d) $v \cdot w \approx 0$;

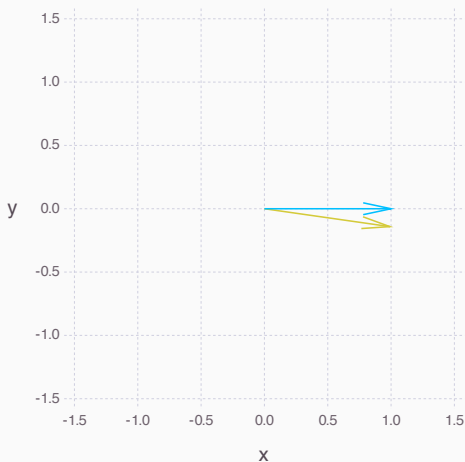
(e) $v \cdot w = 1$;

(f) $v \cdot w \approx 1$;

(g) $v \cdot w = -1$;

(h) $v \cdot w \approx -1$.

Vectors on the unit circle in $\mathbb{R}^2$: true or false?



(a) $v \cdot w > 0$;

(b) $v \cdot w < 0$;

(c) $v \cdot w = 0$;

(d) $v \cdot w \approx 0$;

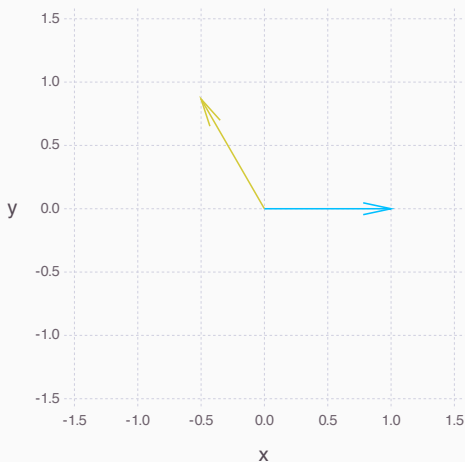
(e) $v \cdot w = 1$;

(f) $v \cdot w \approx 1$;

(g) $v \cdot w = -1$;

(h) $v \cdot w \approx -1$.

Vectors on the unit circle in $\mathbb{R}^2$: true or false?



(a) $v \cdot w > 0$;

(b) $v \cdot w < 0$;

(c) $v \cdot w = 0$;

(d) $v \cdot w \approx 0$;

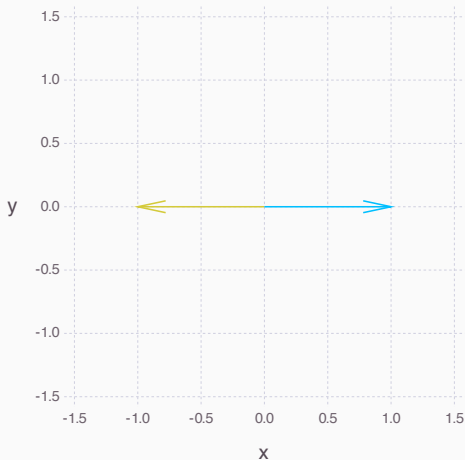
(e) $v \cdot w = 1$;

(f) $v \cdot w \approx 1$;

(g) $v \cdot w = -1$;

(h) $v \cdot w \approx -1$.

Vectors on the unit circle in $\mathbb{R}^2$: true or false?



(a) $v \cdot w > 0$;

(b) $v \cdot w < 0$;

(c) $v \cdot w = 0$;

(d) $v \cdot w \approx 0$;

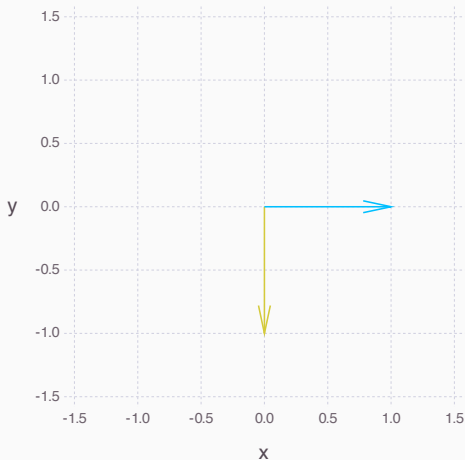
(e) $v \cdot w = 1$;

(f) $v \cdot w \approx 1$;

(g) $v \cdot w = -1$;

(h) $v \cdot w \approx -1$.

Vectors on the unit circle in $\mathbb{R}^2$: true or false?



(a) $v \cdot w > 0$;

(b) $v \cdot w < 0$;

(c) $v \cdot w = 0$;

(d) $v \cdot w \approx 0$;

(e) $v \cdot w = 1$;

(f) $v \cdot w \approx 1$;

(g) $v \cdot w = -1$;

(h) $v \cdot w \approx -1$.

Length of a vector

Definition 1.2.2 of Johnston (2021)

The *length* of a vector $v = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$, denoted by $\|v\|$, is the quantity

$$\|v\| := \sqrt{v \cdot v} = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}.$$

$\|v\|$ is also called the Euclidean norm of the vector v .

Theorem 1.2.2 of Johnston (2021)

Suppose $v \in \mathbb{R}^n$ and $c \in \mathbb{R}$. Then the following properties hold:

- (a) $\|cv\| = |c|\|v\|$;
- (b) $\|v\| \geq 0$ with equality if and only if $v = 0$.

- Vectors with length equal to 1 are called *unit vectors*.
- Any vector $v \neq 0$ can be *normalised* to have length equal to 1.
- We can write every non-zero vector $v \in \mathbb{R}^n$ in the form

$$v = \|v\|u,$$

where $u = v/\|v\|$ is the unique unit vector pointing in the same direction as v .

Renormalisation of a vector in $\mathbb{R}^2$ and the unit circle in $\mathbb{R}^2$

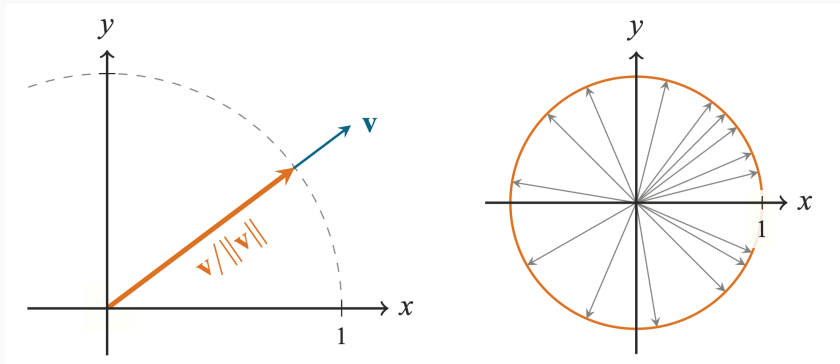


Figure 1.9 of Johnston (2021)

Euclidean vector spaces

Bases

The standard basis

- The set of vectors $\{e_1, e_2, \dots, e_n\}$, where

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots, \quad e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix},$$

is called the *standard basis* in $\mathbb{R}^n$.

- Any vector $v \in \mathbb{R}^n$ can be written in terms of the vectors $e_1, e_2, \dots, e_n$ in the following way

$$v = v_1 e_1 + v_2 e_2 + \dots + v_n e_n. \quad (2)$$

- The standard basis vectors are sometimes denoted by $\hat{i}$, $\hat{j}$, and $\hat{k}$ in $\mathbb{R}^3$.

Euclidean vector spaces

Linear transformations

Functions and transformations

- Recall that a *function* is a rule that associates with each element of a set A one and only one element in a set B and we write

$$f: A \rightarrow B.$$

- If f associates the element b with the element a , then we write

$$b = f(a)$$

and we say that b is the image of a under f or that $f(a)$ is the value of f at a .

- The set A is called the *domain* of f and the set B the *codomain* of f .
- The subset of the codomain that consists of all images of elements in the domain is called the *range* of f .

Section 1.8 of Anton, Rorres, and Kaul (2019)

Functions and transformations (cont.)

- In many applications the domain and codomain of a function are sets of real numbers.
- We will be concerned with functions for which the domain is $\mathbb{R}^n$ and the codomain is $\mathbb{R}^m$ for some positive integers m and n .
- In this setting it is common to use italicised capital letters for functions, the letter T being typical.

Section 1.8 of Anton, Rorres, and Kaul (2019)

Definition 1 on page 78 of Anton, Rorres, and Kaul (2019)

If T is a function with domain $\mathbb{R}^n$ and codomain $\mathbb{R}^m$, then we say that T is an *operator* or a *transformation* from $\mathbb{R}^n$ to $\mathbb{R}^m$ or that T *maps* from $\mathbb{R}^n$ to $\mathbb{R}^m$, which we denote by writing

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

Matrices as transformations

- Consider now an $m \times n$ -matrix A .
- For every vector $x \in \mathbb{R}^n$, the product

$$Ax = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + \dots + a_{1n}x_n \\ a_{21}x_1 + \dots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n \end{bmatrix}$$

is a vector in $\mathbb{R}^m$.

- So we can think of A as a map $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by $T_A(x) := Ax$.
- T_A is called the *matrix operator* or *matrix transformation* associated with A .
- Recommended to watch *Linear transformations and matrices* of 3Blue1Brown.

Properties of matrix transformations

Theorem 1.8.1 of Anton, Rorres, and Kaul (2019)

For $A \in \mathbb{R}^{m \times n}$, $x, y \in \mathbb{R}^n$ and $k \in \mathbb{R}$, the following holds

- (1) $T_A(0) = 0$;
- (2) $T_A(k \cdot x) = k \cdot T_A(x)$ (homogeneity);
- (3) $T_A(x + y) = T_A(x) + T_A(y)$ (additivity);
- (4) $T_A(x - y) = T_A(x) - T_A(y)$.

- All these properties follow from our matrix multiplication rules.
Note that (2) and (3) together imply (1) and (4).

Linear transformation

Given some map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, how can we know if it is a matrix transformation? To answer this question, we need to learn what a linear transformation is.

Definition 1.4.1 of Johnston (2021)

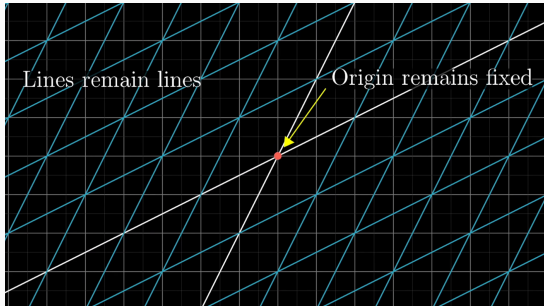
A *linear transformation* is a function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ that satisfies the following two properties:

- (a) $T(v + w) = T(v) + T(w)$ for all vectors $v, w \in \mathbb{R}^n$;
- (b) $T(cv) = cT(v)$ for all vectors $v \in \mathbb{R}^n$ and all scalars $c \in \mathbb{R}$.

Intuition

Visually speaking, a transformation is “linear” if it has two properties

- (i) all lines must remain lines, without getting curved;
- (ii) the origin must remain fixed in place.



Linear transformations and matrices by 3Blue1Brown

References



Johnston, Nathaniel (2021). *Introduction to linear and matrix algebra*. Springer Cham.



Anton, Howard, Chris Rorres, and Anton Kaul (Sept. 2019). *Elementary linear algebra: applications version*. 12th edition. John Wiley & Sons, Inc.